Alexander Greif

alex@alexgreif.de

portfolio.alexgreif.de github.com/alexgreif linkedin.com/in/alexgreif



Profile

I work in statistical consulting and tutoring, with previous experience in university research and IT teaching and a background in sports science. My ongoing studies in mathematical-technical software development complement practical experience in data analysis, programming and technical communication.

Professional Experience

Statistical Consultant - Self-employed — Remote
Working with direct private clients and for statistical consultancy NOVUSTAT — Apr 2023 – present

- Provide methodological and statistical advice primarily for bachelor's, master's and doctoral theses, helping clients develop research questions and hypotheses, plan data collection and select statistical methods.
- Conduct data processing and statistical analyses in R, SPSS, and Excel.
- Help clients present methods and findings clearly and accurately in their academic work.

Tutor in Statistics, Mathematics and Computer Science - Self-employed — Remote
Direct private clients; previously also for Studienkreis — Apr 2023 – present

- Provide tutoring for university students in statistics, applied mathematics, programming and databases.

Freelance IT Instructor - Qualidy — Remote
Teaching at bfz in vocational retraining programmes for adults to become IT specialists in application development or system integration — Apr 2026 – Sept 2026

- Planned and delivered one-week online modules covering the fundamentals of Python and C# programming, Windows, Linux, Docker, Arduino and IT service management.

Data Analysis Tutor - DataCraft — Remote
Working student position; daily tutoring throughout a one-year data analyst training programme — Mar 2024 – May 2025

- Supported participants during individual assignments and group projects, helping them apply course content to practical, business-oriented data analysis tasks.
- The course primarily used Python throughout the data analysis workflow: acquiring data via APIs and web scraping, processing data with pandas, visualising data with Matplotlib and Plotly, and developing data applications with Streamlit and FastAPI.
- Further topics included SQLite and PostgreSQL databases, Excel and Power BI.

Student Research Assistant - University of Münster — Münster, Germany
Education and Teaching in Sports, Institute of Sport and Exercise Sciences — Apr 2021 – Apr 2023

- Provided methodological and statistical advice across research projects.
- Developed a structured R project integrating data preparation, statistical analysis and R Markdown reporting for a controlled intervention study with repeated measurements, combining data from multiple sources.
- Conducted statistical analyses of cross-sectional survey data and wrote the quantitative methods and results sections of the resulting co-authored article on gender differences in student satisfaction.

Sports Science Intern - Plymouth Argyle FC Academy — Plymouth, United Kingdom
Erasmus-supported university placement — Aug 2019 – Dec 2019

- Analysed GPS and heart-rate data in Excel to track changes in player workload and develop physical demand profiles for training drills to support training planning.
- Planned and led on-field and gym-based strength and conditioning sessions.

Education

FernUniversität in Hagen — Germany
Mathematical-Technical Software Development - B.Sc. programme, ongoing — Oct 2022 – present
Part-time distance study; 160/180 ECTS completed; average grade: 1.9 (German scale)

- Mathematical and statistical foundations covering probability theory, measure and integration theory, parametric statistics, linear algebra, real analysis and numerical methods.
- Computer science and software development covering data structures and algorithms, object-oriented programming, relational databases and software engineering.

University of Münster — Münster, Germany
Sports, Exercise and Human Performance - M.Sc. programme, discontinued — Oct 2020 – Mar 2023
Taught and assessed in English; discontinued without completing the degree

- Collected and analysed biomechanical and neurophysiological data, including kinematic and kinetic measurements, EMG and EEG; applied computational models in MATLAB to estimate joint moments and internal forces.

University of Münster — Münster, Germany
Human Movement in Sports and Exercise - B.Sc. — Oct 2017 – Mar 2021
Taught and assessed in English; final grade: 1.8 (German scale)

- Coursework covering biomechanics, motor control, exercise physiology, sport psychology and social aspects of sport.
- Training in study design, statistics and scientific communication, including practical experience through coursework projects.

Technical Skills

Programming: R, Python, Java, SQL; basic knowledge of C, C# and Bash

Data Analysis and Reporting: tidyverse, ggplot2, pandas, NumPy, SPSS, Excel, R Markdown

Additional IT Skills: Git, GitHub, Docker, Linux

Selected Projects

The integration of external and internal match load with contextual factors to assess training status in football — Mar 2021
Bachelor's thesis, University of Münster; thesis grade: 1.3 (German scale)

- Processed and combined GPS, heart-rate and match-context data from a football team's matches over an 18-month period.
- Developed and compared multilevel models of mean match heart rate in R and examined deviations from model predictions as a potential fatigue indicator.

Anguilla Search — Jan 2025
Programming practical assignment, FernUniversität in Hagen

- Built a Java search engine with web crawling and scraping, text processing and indexing.
- Implemented search ranking with TF-IDF, cosine similarity and PageRank.
- Added an interactive command-line interface and implemented JUnit tests.

Survival Analysis — May 2025
Statistical practical assignment in Mathematical Statistics, FernUniversität in Hagen

- Analysed right-censored survival data in R using Kaplan–Meier curves and a log-rank test to compare treatment and placebo groups.
- Estimated statistical power and the number of events needed for 80% power.
- Compared age distributions between groups using the Wilcoxon rank-sum test to investigate age differences as a potential confounding factor.

Cyclistic Case Study Sept 2022

Capstone project, Google Data Analytics Professional Certificate

- Processed and cleaned 5.7 million bike-share trip records in R, documenting decisions and their rationale.
- Compared temporal and geographic usage patterns of casual riders and members.
- Produced visualisations, R Markdown reports and a presentation linking findings to recommendations for converting casual riders into members.

Publications

Magner, A., **Greif, A.**, & Kohake, K. (2026). Studienzufriedenheit im Sportlehramtsstudium: Eine Mixed-Methods-Studie unter besonderer Berücksichtigung des Geschlechts [Student satisfaction in physical education teacher education: A mixed-methods study with particular attention to gender]. *Beiträge zur Hochschulforschung*, *48*(2), 110–139. https://doi.org/10.58069/bzh.2026.2.192

Professional Development

Google Data Analytics Professional Certificate Sept 2022

Data preparation, analysis and visualisation using Google Sheets, SQL and R

Languages

German: Native

English: C1 (CEFR, self-assessed)

Spanish: B1 (CEFR, self-assessed)